Modeling Memory Consolidation and Representational Drift: A Computational Investigation of Hippocampal Map Plasticity via Continual Learning

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Background & Introduction

The hippocampus supports spatial navigation via a "cognitive map" of place cells. Paradoxically, while this map is stable for navigation, the underlying neural code for any given location exhibits "representational drift," gradually changing over days..

A recent landmark study by Geva, Deitch, Rubin, and Ziv (2023) sought to deconstruct the factors driving this phenomenon. By longitudinally imaging CA1 neurons in mice exploring two familiar environments at different time intervals, they performed a key dissociation between the effects of time and experience. **Their findings revealed a double dissociation:**

1. The passage of time was the dominant factor driving changes in the overall activity rates of neurons (i.e.
2. The amount of active experience within an environment was the dominant factor driving changes in the spatial tuning of the cells (i.e., the location of their place fields).

This suggests that **representational drift is not a unitary process** but a multi-faceted phenomenon governed by distinct biological mechanisms. The experience-dependent nature of spatial tuning drift points towards an active, plasticity-driven process of memory updating and reconsolidation.

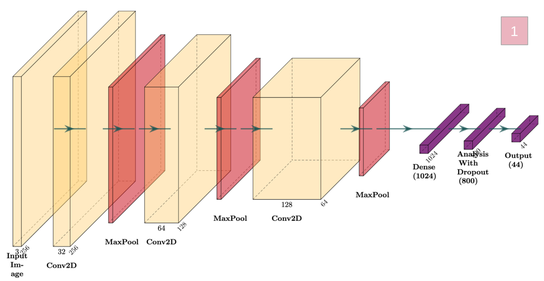
Inspired by these findings, this computational project aims to investigate the underlying mechanisms of experience-dependent drift. We will train a deep Convolutional Neural Network (CNN) on first-person video to simulate this learning process. By iteratively fine-tuning the network on augmented data to simulate daily experience, we aim to computationally replicate the drift observed by Geva et al. and explore the hypothesis that drift is a functional feature for maintaining a flexible and adaptive cognitive map.

Methodology

1. **Data Gathering & Cleaning:** To generate realistic visual input, 10-second, 30 FPS videos were filmed from a horizontally stabilized push-cart. All video frames were subsequently resized to 256x256 pixels for input into the convolutional neural network.
2. **Model Architecture:**

- The model is a CNN performing a multi-label ordinal regression task, mapping 256x256 RGB images to a 22-unit vector representing cumulative progress.
- The architecture uses three Conv2D/MaxPooling2D blocks that feed into dropout and dense layers; a sigmoid output layer predicts the ordinal position.

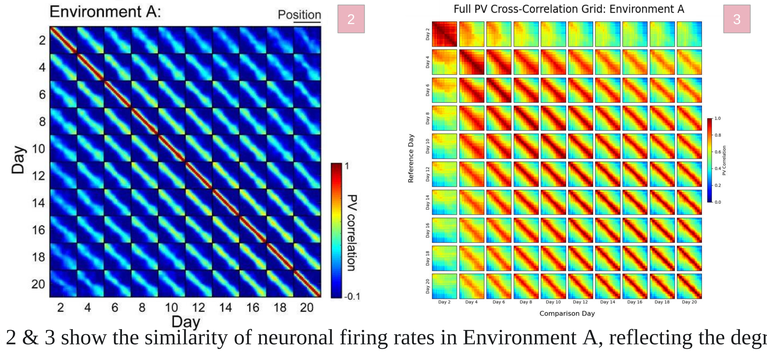
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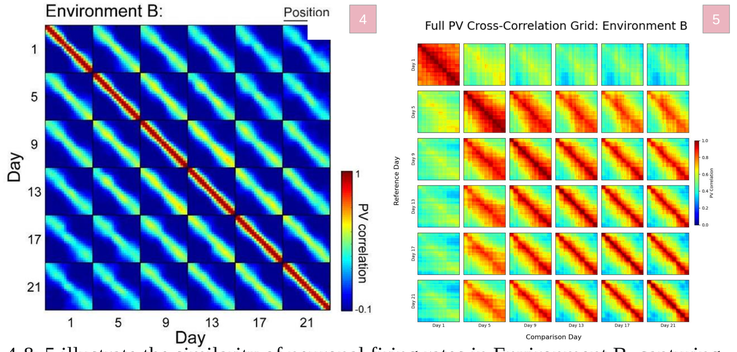
3. **Simulating Drift:** The model was trained over a 21-day schedule alternating between familiar (A), novel (B), and mixed-rehearsal environments, with nightly fine-tuning on random data subsets using a high learning rate to simulate plasticity and memory drift. Following Lu, et al. (2023), we calculated Tuning Curve, Population Vector, and Ensemble Rate correlations, then plotted the averaged scores against time differences to generate memory decay curves.

Results

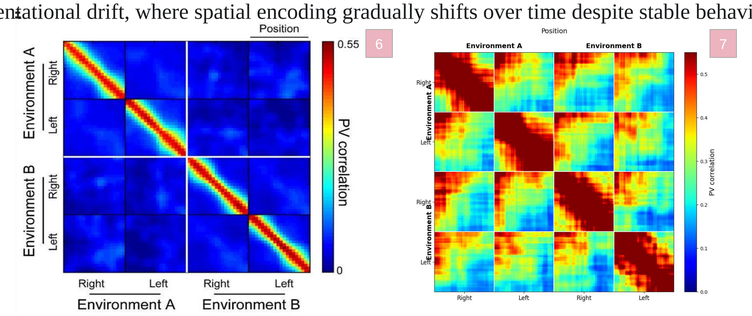
Paper's Results | **Our Results**



Figures 2 & 3 show the similarity of neuronal firing rates in Environment A, reflecting the degree to which specific locations were revisited. Correlations are strongest along the diagonal, as neurons were sorted by activity on the reference day, but decline with increasing temporal distance between days. This reduction indicates representational drift, where spatial representations gradually shift over time despite consistent experience.



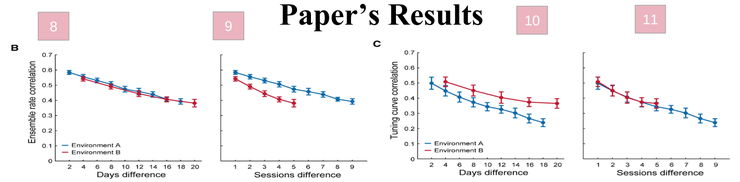
Figures 4 & 5 illustrate the similarity of neuronal firing rates in Environment B, capturing temporal dynamics across days. Correlations peak along the diagonal, reflecting sorting by the reference day, but weaken as the interval between sessions increases. This decline signifies representational drift, where spatial encoding gradually shifts over time despite stable behavior.



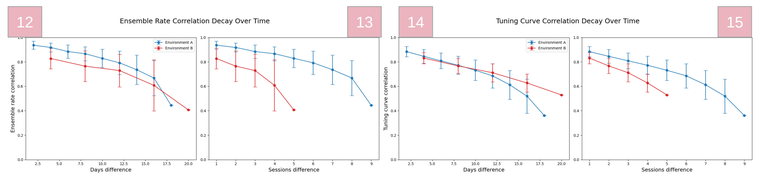
Figures 6 & 7 provide a summary of the preceding PV correlation analyses, presenting a population vector that integrates spatial location, running direction, and environmental context, averaged across all session pairs.

Conclusion

Paper's Results



Our Results



Figures 12 and 13 display the ensemble rate correlation from our simulation which aligns with Figures 8 and 9 from the paper. From these graphs we are able to align our findings with the ones in the paper in which experience (sessions) affect spatial changes more than temporal changes. We hypothesized that because environment B had longer days between the addition of new data in addition to dropout layers, the neurons have a much higher chance of adjusting its weights and thus leading to a much more drastic change in the PV correlation over a shorter amount of time. Figure 14 and 15 correspond to the tuning curve of the paper's graph displayed on 10 and 11. However, graph 15 seems to deviate in response to the environments more than graph 11. We hypothesized the addition of the dropout layer to be part of the reason for the differences in our tuning curves for Env. A and Env. B; biological neurons can't zero out like our artificial neurons – they adjust gradually.

Future Plans

- Refine the model to more closely replicate the findings reported in the biological study.
- Extract and analyze data from the model to identify patterns that could provide a more comprehensive explanation of the results.
- Examine the raw neural data from the biological study to determine whether the patterns observed in the artificial model are also present in biological systems.

Acknowledgements

Geva, N., Deitch, D., Rubin, A., & Ziv, Y. (2023). Time and experience differentially affect distinct aspects of hippocampal representational drift. Neuron, 111(14), 2241–2253. https://doi.org/10.1016/j.neuron.2023.05.005